Bridging Land and Sky: STELLA as a Tool of Plant Stress Analysis and Increasing Spatial Resolution of Satellite Spectroscopy Measurements

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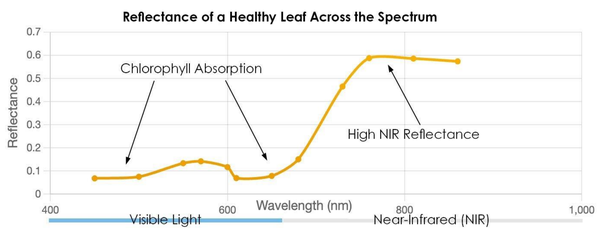
What is STELLA?

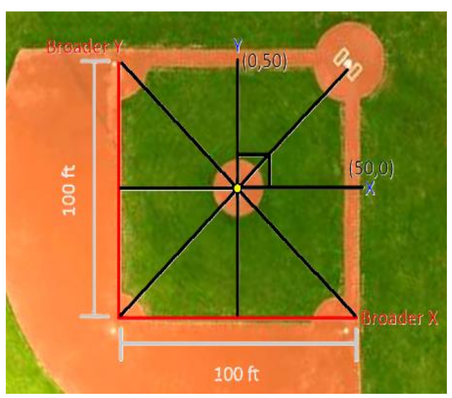
STELLA is a handheld spectrometer equipped to record irradiance data across the Visible and Near Infrared spectrum. Because pigments within plants follow patterns of absorbance and reflectance values, CARMA uses STELLA to predict the condition of such pigments, becoming an invaluable tool in the team's ecological research. With STELLA, CARMA can conveniently monitor individual plant health at a molecular level.

The STELLA 1.1 Device (NASA)



(NASA)

From Baseball Fields to Blades of Grass



A baseball field (the approximate size of one pixel in LANDSAT) overlayed with CARMA's experimental set-up.

The CARMA team recording data with STELLA at Rafter W. Ranch

LANDSAT imagery cannot identify acute ecological trends. This is due to the observational nature of LANDSAT's Visible and Near Infrared lens, as each data point (or pixel) is representative of a 30m x 30m square. STELLA strengthens LANDSAT imagery by providing spectral signatures of individual plants, contextualizing broad-scale processes on a crop-by-crop basis.

Filtering Out the Noise: Data Analysis

Indices quantify plant health without dependance on prior documentation of species-specific imaging. They are primarily useful in correcting for environmental influences which may affect STELLA's readings.

- SAVI: minimizes soil effects and measures vegetation greenness (where "L" is the soil adjustment factor)
- NDVI: simple but sensitive to soil and atmospheric effects. Measures vegetation greenness on a –1 to 1 scale, utilizing the fact that healthy plants reflect NIR light and absorb red light.

$$NDVI = \frac{NIR - RED}{NIR + RED} \qquad SAVI = \frac{(NIR - R)(1+L)}{(NIR + R + L)}$$

STELLA and CONSERVATION

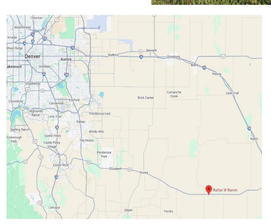
In a changing climate, ground-based spectroscopy helps inform local environmentalists about the impacts of conservational efforts.

Rafter W. Ranch: Rancher Lance Wheeler sought an efficient way to monitor the health of the plant life on his pasture, which is crucial to the health of his livestock. CARMA uses STELLA to take readings throughout Lance's property and identify at-risk species.

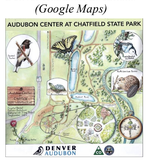
Audubon: At the Denver Audobon Kingery Nature Center, the maintenance of native plant life is essential to maintain the natural migratory patterns of birds. Their conservation efforts benefit from the time-effective data STELLA provides.

A photo taken of rancher Lance Wheeler on his Property (photo credit: Farmranchhomestead.com)

Denver Audubon Locations (Google Maps)



Location of Rafter W. Ranch (Google Maps)



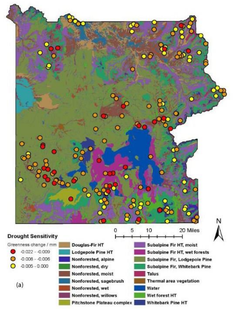
Illustrated Map of the Kingery Nature Center

Rosa arkansana (prairie rose) taken from Rafter W. Ranch

Applications of Handheld Spectroscopy

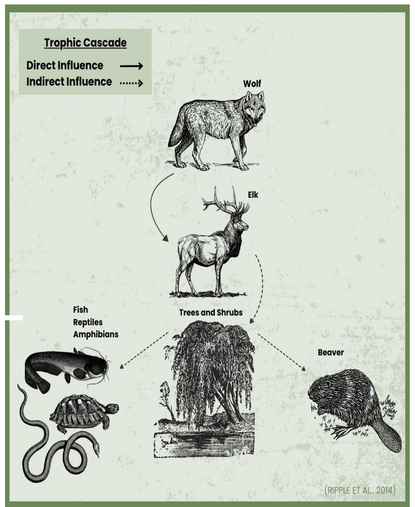
The CARMA team is currently working on establishing STELLA within the field of ecological research. CARMA has plans to explore three diverse areas of research within Yellowstone National Park. CARMA will,

- Record the state of aspen trees following wolf reintroduction and the subsequent trophic cascades such an ecological shift brings.
- Compare spectral readings of Yellowstone's wetland plants to prior data collected on drought conditions within the park in order to contribute to current park monitoring efforts.
- Compare spectral readings of *D. Lanuginosum,* given an active symbiosis between it and the fungus *curvularia*, with other wetland plants to determine how symbiosis effects drought resilience.



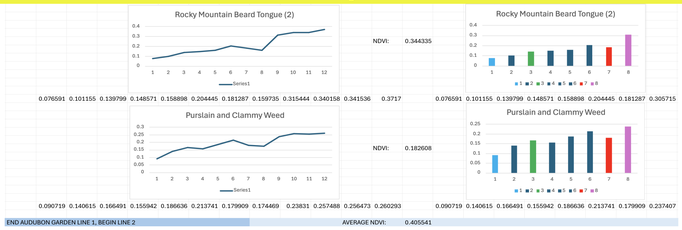
Map of Drought-Sensitive Areas in Yellowstone National Park (Thoma, et al.)

D. Lanuginosum, recorded to grow at 46.5 degrees Celsius (National Park Service)



Above, a diagram depicts the direct and indirect influences wolves have on ecosystems.

Preliminary Results



A screenshot of CARMA's analysis of data collected with the STELLA at Denver Audubon's Kingery Nature Center.

References

Thoma, D., et al. (2021, June 8). *Patterns of primary production & ecological drought in Yellowstone (U.S. National Park Service)*. National Parks Service. https://www.nps.gov/articles/patterns-of-primary-production-ecological-drought-in-yellowstone.htm
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